

Resistivity

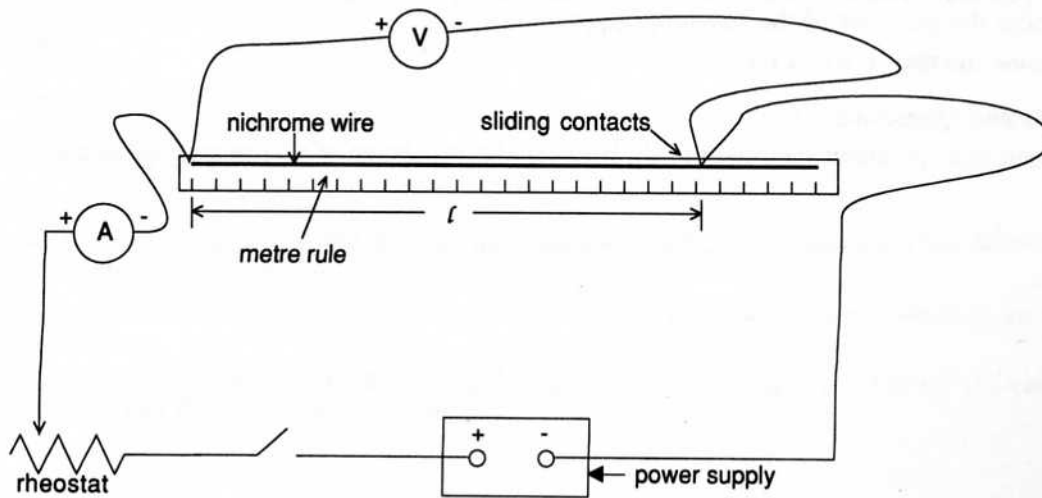
The resistivity of a conductor of length ℓ metres, cross-sectional area A square metres, and resistance R ohms is:

$$\rho = \frac{RA}{\ell}$$

Resistivity is a characteristic of a conductor.

You can determine the resistance of the section of the wire in the circuit diagram below using:

$$R = \rho \frac{\ell}{A}$$



this experiment you will calculate the resistivity of a piece of thin wire.

Equipment needed

nichrome wire (1 m)
ammeter (0-5 A)
voltmeter (0-12 V)
rheostat
sliding contact
power supply (0-12 V DC)
switch
six electrical leads
metre rule
micrometer screw gauge

Procedure

- 1 Set up the apparatus as shown in the diagram above.
- 2 Use a micrometer screw gauge to measure the diameter of the nichrome wire five times. Ask for help if you do not know how to use a micrometer screw gauge. Assume the wire has a uniform cross-section.
- 3 Calculate the mean diameter of the nichrome wire.
- 4 Calculate the cross-sectional area of the nichrome wire.
- 5 Set the power supply to 6 V and switch on your circuit.
- 6 Move the sliding contact so the current flows through 0.80 m of the nichrome wire. Record the potential difference across l , and the current flowing in the circuit.
- 7 Repeat step 6 for lengths of nichrome of 0.60, 0.40 and 0.20 m.

Processing the results

- 1 Copy and complete the following table:

l = Wire length (m)	potential difference (V)	Current (A)	Resistance of wire (Ω)	$\frac{R}{l}$
			Mean =	

- 2 Determine the mean value for the ratio of the resistance to the length, taking care with the accuracy you claim for your results.
Note: You may want to obtain the mean of the ratio by plotting the resistance against length to determine the gradient of the resulting graph.
- 3 Determine the resistivity of the wire.

Conclusions and Questions

- 1 What can you say about the relationship between the resistance of a wire and its length?